

```
1 import java.util.*;
2 class RotateMatrix
3 {
4     int a[][][],m;
5
6     void input()
7     {
8         Scanner scan=new Scanner(System.in);
9         System.out.println("Enter the number of rows (greater than 2 and less than
10) for the square matrix : ");
10        m=scan.nextInt();
11
12        if(m<=2 || m>=10)
13            System.out.println("SIZE OUT OF RANGE");
14
15        else{
16            a=new int[m][m];
17            int i,j;
18
19            System.out.println("Enter "+(m*m)+" numbers for the matrix: ");
20            for(i=0;i < m;i++){
21                for(j=0;j < m;j++){
22                    a[i][j] = scan.nextInt();
23                }//loop j
24            }//loop i
25        }
26    }
27
28    RotateMatrix rotate(RotateMatrix matrix)
29    {
30        RotateMatrix k=matrix;
31        for (int i = 0; i < m; i++)
32        {
33            for (int j = i; j < m; j++)
34            {
35                int temp= k.a[i][j];
36                k.a[i][j]= k.a[j][i];
37                k.a[j][i]= temp;
38            }
39        }
40        for(int i=0;i<m;i++)
41        {
42            int top=0;
43            int bottom = m-1;
44            while(top<bottom)
45            {
46                int temp = k.a[top][i];
47                k.a[top][i]=k.a[bottom][i];
48                k.a[bottom][i] = temp;
49                top++;
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50         bottom--;
51     }
52 }
53 System.out.println("THE ROTATED MATRIX IS ");
54 return k;
55
56 }
57
58 void display()
59 {
60     for(int i=0;i < m;i++){
61         for(int j=0;j < m;j++){
62             System.out.print(a[i][j] + " ");
63         }//loop j
64         System.out.println();
65     }//loop i
66     System.out.println();
67 }
68
69 void MaxMin(RotateMatrix r)
70 {
71     int max=0,min=a[0][0];
72     for(int i=0;i<m;i++)
73     {
74         for(int j=0;j<m;j++)
75         {
76             if(r.a[i][j]>max)
77             {
78                 max=r.a[i][j];
79             }
80
81
82             if(r.a[i][j]<min)
83             {
84                 min=r.a[i][j];
85             }
86         }
87     }
88
89     System.out.println("THE MAXIMUM ELEMNT :"+max);
90     System.out.println("THE MINIMUM ELEMNT :"+min);
91 }
92
93
94 public static void main()
95 {
96     RotateMatrix ob=new RotateMatrix();
97     ob.input();
98     ob.display();
99     RotateMatrix obj=new RotateMatrix();
```

```
100         obj=obj.rotate(ob);
101         obj.display();
102         ob.MaxMin(ob);
103     }
104 }
105
```